



Society of Petroleum Engineers

SPE-229646-MS

Large Language Model Empowered Automated Reservoir Agent: A Win-Win Strategy for Reservoir Intelligent Management and Transformation

Xing Zhao, Zhenhua Oil Co., Ltd, Beijing, China / China Zhenhua Oil, HONG KONG Co., Ltd, HK, China; Shuwu Wei, Chengdu North Petroleum Exploration and Development TechnologyCo., Ltd, Chengdu, Sichuan, China; Lian Wang, Zhenhua Oil Co., Ltd, Beijing, China / China Zhenhua Oil, HONG KONG Co., Ltd, HK, China; Rui Deng, Zhenhua Oil Co., Ltd, Beijing, China / Chengdu North Petroleum Exploration and Development TechnologyCo., Ltd, Chengdu, Sichuan, China; Liang Zhang, Zhenhua Oil Co., Ltd, Beijing, China / China Zhenhua Oil, HONG KONG Co., Ltd, HK, China; Dinghao Liu and Lingfeng Yang, Chengdu North Petroleum Exploration and Development TechnologyCo., Ltd, Chengdu, Sichuan, China; Keyi Zang, Zhenhua Oil Co., Ltd, Beijing, China / China Zhenhua Oil, HONG KONG Co., Ltd, HK, China; Yong Deng, Zhenhua Oil Co., Ltd, Beijing, China / China Zhenhua Oil, HONG KONG Co., Ltd, HK, China; Jiafeng Xu, Zhenhua Oil Co., Ltd, Beijing, China / China Zhenhua Oil, HONG KONG Co., Ltd, HK, China; Yang Luo, Zhenhua Oil Co., Ltd, Beijing, China

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229646-MS](https://doi.org/10.2118/229646-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

The upstream oil and gas industry is currently undergoing digital transformation, presenting numerous opportunities as well as significant challenges, particularly in the digitalization of reservoir management with Large Language Model (LLM). Meanwhile, reservoir management is a field rich in specialized knowledge and technical complexity. Most of the LLMs available today are designed for general domains and lack a deep understanding of specialized knowledge, resulting in suboptimal performance in the realm of automated reservoir agents. To address this issue, we introduce the Knowledge-Retrieval–DeepSeek System (KRDS), an LLM-powered automated reservoir agent that integrates a two-stage knowledge retriever with a retrieval-augmented conversational DeepSeek LLM deployed on-premises to ensure data sovereignty.

The KRDS automated reservoir agent consists of two components: the two-stage knowledge retrieval and recall model and the retrieval-augmented conversational DeepSeek LLM framework. By injecting the top-ranked passages returned by the retriever, the LLM acquires reservoir-engineering context that is otherwise absent from general-purpose models. Then the retrieval-augmented conversational DeepSeek LLM framework invokes tool functions to retrieve reservoir properties, production history, future conditions, and more, in order to provide feedback on the query of reservoir engineer. In this way, the automated reservoir agent calls the relevant tool functions in the system based on the query, completes reservoir management tasks, and presents the results to the reservoir engineer in a question-and-answer format.

To verify the effectiveness and reliability of the KRDS automated reservoir agents, the intelligent management of a typical carbonate reservoir in the Persian Gulf Basin of the Middle East is

used as an example. The establishment and application of KRDS automated reservoir agents consists of four sections: dataset, evaluation index, experimental facility, and main procedures. KRDS is furnished with eighteen specialized reservoir management tool functions including 'get_permeability,' 'get_porosity,' 'get_production_rate,' 'well trajectory,' and others, tasked to answer forty-five scenario-based questions reflecting daily operational practice. For the query 'Check the production history of A1 well,' KRDS automatically renders oil-, water- and injection-rate curves, achieving the goal of utilizing the LLM to empower the automated reservoir agent for reservoir intelligent management and transformation. Quantitatively, the success rate and correct path per hundred experiments are 83% and 87%, respectively.

To the best of our knowledge, KRDS represents the first domain-tailored LLM agent that autonomously executes reservoir-management tasks while maintaining enterprise-level data security. The results demonstrate that coupling targeted retrieval with an open-access LLM offers a feasible pathway toward intelligent reservoir management and digital transformation in the upstream sector.

Keywords: Automated Reservoir Agent, Large Language Model, Reservoir Intelligent Management and Transformation, Middle East Carbonate Reservoir, DeepSeek

Introduction

Large-scale pre-trained models are now at the vanguard of a new era of technological and industrial transformation, with their application in specific domains emerging as a pivotal trajectory for future intelligent solutions (Liu et al., 2020). In mature oilfields worldwide, reservoir engineers are confronted with an immense workload, a predicament stemming from the proliferation of wells and the intricate nature of producing layers (Parshall, 2012). Consequently, the development of AI agents specifically tailored for reservoir dynamic analysis has become an imperative, as these systems possess the capacity to significantly enhance operational efficiency by automating a range of critical functions, including well history data analysis, cross-well section plotting, and the calculation of key technical indicators (Mohaghegh, 2011). This innovative approach promises to displace a substantial portion of manual labor through intuitive human-machine dialogue, thereby streamlining the development process for oil and gas fields (Wang et al., 2023). While general-purpose large language models (LLMs), such as OpenAI's GPT, Microsoft's PaLM, and open-source models like LLaMA, have demonstrated remarkable capabilities in language comprehension and complex task execution (Wolf et al., 2020; Osco et al., 2023; Chang et al., 2024), their direct application in highly specialized fields like oil and gas exploration and development presents distinct and formidable challenges.

A paramount challenge impeding the effective digitalization of reservoir management through LLMs is their inherent lack of domain-specific expertise. As perceptively highlighted by Liu et al. (2024), these general-purpose models typically lack the deep understanding of technical terminology and specialized knowledge essential for rigorous reservoir engineering. This fundamental deficiency severely curtails their direct utility in the intricate and demanding realm of oil and gas exploration and development. As a result, the strategic focus for LLM implementation within this industry must necessarily pivot from the creation of general-purpose foundational models towards the meticulous fine-tuning and adaptation of existing models for specific downstream tasks, all of which must be rigorously guided by practical business imperatives.

Reservoir management is a domain characterized by its knowledge-intensive nature and profound technical complexity. Regrettably, most contemporary LLMs are architected for broad, general-purpose applications and consequently lack the requisite specialized knowledge to function as effective automated reservoir agents, a shortcoming that invariably leads to suboptimal performance. Therefore, a compelling need exists for a domain-tailored LLM agent capable of autonomously executing intricate reservoir management tasks while simultaneously upholding the rigorous standards of enterprise-level data security.

To effectively bridge this critical gap, we introduce the Knowledge-Retrieval–DeepSeek System (KRDS), an LLM-powered automated reservoir agent engineered to address these challenges. The KRDS architecture seamlessly integrates a sophisticated two-stage knowledge retrieval mechanism with a retrieval-augmented conversational DeepSeek LLM, which is strategically deployed on-premises to guarantee data sovereignty and security. The system's efficacy is predicated on two core components: a two-stage knowledge retrieval and recall model and a retrieval-augmented conversational DeepSeek LLM framework. This framework augments the LLM with specialized reservoir-engineering context by systematically injecting relevant passages retrieved from a comprehensive knowledge base. Subsequently, it can intelligently invoke a suite of specialized tool functions to access and analyze critical data on reservoir properties and production history, thereby providing engineers with comprehensive and data-driven feedback to their complex queries.

The remainder of this paper is structured as follows: Section 2 delineates the related work of the KRDS system. Section 3 provides an in-depth description of the framework of KRDS. Section 4 presents the case study. Finally, Section 5 offers our conclusions and outlines potential avenues for future research.

Related work

This section details the comprehensive architecture and methodology of the proposed Knowledge-Retrieval–DeepSeek System (KRDS), engineered to address the unique challenges of intelligent reservoir management. The KRDS framework is built upon three foundational pillars: a sophisticated two-stage knowledge retrieval and recall mechanism, a strategically deployed DeepSeek large language model for efficient technical reasoning and data sovereignty, and an innovative retrieval-augmented conversational framework that empowers the agent with specialized tool-use capabilities. The following subsections elaborate on each of these components, explaining their individual roles and how they are integrated to create a robust and autonomous agent for comprehensive reservoir dynamic analysis.

Two-stage knowledge retrieval and recall

Open-domain QA systems now routinely adopt a two-stage pipeline in which a lightweight *recall* module maximises coverage and a heavyweight *reranker* refines relevance. Dense Passage Retrieval (DPR) remains the canonical recall model, offering order-of-magnitude speed-ups over exhaustive search without sacrificing recall (Karpukhin et al., 2020). The architecture leverages dual-encoder networks to separately encode queries and passages into dense vectors, enabling efficient similarity search through approximate nearest neighbor algorithms. This approach has proven particularly effective in handling large-scale petroleum engineering documentation, where traditional keyword-based methods often fail to capture semantic relationships between technical concepts.

Hybrid recall that blends sparse (e.g., BM25) and dense signals further lifts recall in technical corpora, as demonstrated by Zhang et al. (2024). Their work reveals that sparse retrieval excels at exact terminology matching—crucial for reservoir-specific nomenclature like "water-oil contact" or "bubble point pressure"—while dense retrieval captures semantic similarity across paraphrases and related concepts. The complementary nature of these approaches becomes especially apparent in multilingual petroleum operations where the same geological formation might be described using different technical terms across regions.

For reranking, cross-encoders such as the BERT reranker proposed by Nogueira and Cho (2019) jointly encode query-passage pairs, enabling fine-grained relevance modeling that captures subtle contextual dependencies. Domain-tuned models like bge-reranker further enhance performance by incorporating technical domain knowledge during pre-training, achieving significant improvements over general-purpose rerankers when processing reservoir engineering documents. These models demonstrate particular strength

in distinguishing between superficially similar but technically distinct concepts, such as differentiating between "effective porosity" and "total porosity" based on contextual clues.

In multilingual oil-and-gas workflows, the BCEmbedding framework achieves state-of-the-art bilingual retrieval, underscoring the value of specialised embeddings for reservoir documentation (NetEase, 2023). The framework's ability to align technical terminology across languages proves invaluable in international operations where reservoir reports, drilling logs, and production data may exist in multiple languages. This cross-lingual capability ensures that critical insights are not lost due to language barriers, enabling comprehensive knowledge retrieval across an organization's entire document repository.

DeepSeek LLM

The DeepSeek family illustrates rapid progress in open-source LLMs oriented toward technical reasoning. DeepSeek-V2 introduces Multi-head Latent Attention (MLA), cutting inference cost while matching or exceeding GPT-3.5 on mathematical and code tasks (DeepSeek Team, 2024a). This architectural innovation compresses the key-value cache through low-rank decomposition, reducing memory requirements by up to 95% compared to traditional multi-head attention. For reservoir engineering applications, this efficiency translates to the ability to process longer contexts—such as complete well logs or extensive production histories—without prohibitive computational costs.

DeepSeek-Coder pushes these gains to 67 billion parameters and leads public models on code-generation benchmarks—an asset when automating reservoir-simulation scripts (DeepSeek Team, 2024b). The model's pre-training on 2 trillion tokens of code and technical documentation equips it with deep understanding of programming patterns, numerical methods, and scientific computing workflows. This foundation proves particularly valuable for generating Eclipse simulation decks, automating decline curve analysis, or scripting complex reservoir property calculations that traditionally require manual coding by petroleum engineers.

Enterprise adoption is eased by efficient 4- and 8-bit quantisation; Liu et al. showed <3 pp accuracy loss after quantising DeepSeek for on-prem deployment (Liu et al., 2025). Their comprehensive evaluation across reservoir-specific benchmarks—including material balance calculations, PVT analysis, and well test interpretation—demonstrates that quantized models maintain professional-grade accuracy while reducing infrastructure requirements by 75%. This finding is crucial for oil and gas companies operating in regions with limited cloud connectivity or strict data sovereignty requirements.

Further, parameter-efficient fine-tuning (e.g., LoRA) tailors DeepSeek to reservoir jargon with modest compute budgets. By updating only 0.1% of model parameters, organizations can adapt the base model to understand company-specific terminology, local geological formations, and proprietary analysis methodologies. This customization capability enables DeepSeek to function as a domain expert that speaks the precise technical language of each organization, improving both user acceptance and practical utility in daily reservoir management tasks.

Retrieval-augmented conversational DeepSeek LLM framework

Retrieval-Augmented Generation (RAG) first proved that external documents can be injected into the LM context at inference time to boost factuality (Lewis et al., 2020). The seminal work by Lewis et al. demonstrated that coupling a pre-trained seq2seq model with a dense passage retriever could dramatically reduce hallucination on knowledge-intensive tasks. For reservoir engineering, where accuracy is paramount and errors can lead to costly operational decisions, this grounding in retrieved documentation provides essential safeguards against the generation of plausible but incorrect technical information.

Self-RAG extended this idea with a self-reflective trigger that retrieves only when needed—ideal for multi-turn engineering dialogues (Asai et al., 2023). The framework introduces special reflection tokens that allow the model to explicitly reason about whether retrieval is necessary, what type of information to retrieve, and how to integrate retrieved passages into its response. In reservoir management conversations, this selective retrieval prevents information overload while ensuring that critical technical details are

available when discussing complex topics like enhanced oil recovery mechanisms or multiphase flow behavior.

Tool-augmented agents merge RAG with actionable APIs. The ReAct paradigm combines reasoning traces and tool calls (Yao et al., 2023), enabling models to decompose complex reservoir engineering tasks into executable steps. For instance, when asked to optimize production from a multilayer reservoir, ReAct-based agents can reason through the required analyses, invoke reservoir simulation tools, query production databases, and synthesize results into actionable recommendations. ToolFormer shows that LLMs can learn tool usage from unlabeled traces (Schick et al., 2023), suggesting that reservoir agents can acquire new computational capabilities through exposure to engineer-tool interactions without explicit programming.

Recent frameworks (e.g., LevelRAG) formalise best practices for retrieval quality, context fusion, and response generation in technical domains (Gao et al., 2025). Their hierarchical approach to retrieval—first identifying relevant document categories, then specific passages, and finally extracting precise technical details—mirrors the cognitive process of experienced reservoir engineers. The framework's emphasis on maintaining technical coherence across multiple retrieved sources proves essential when synthesizing insights from diverse data types including seismic interpretations, well logs, core analyses, and production histories.

These advances collectively motivate KRDS: a domain-adapted DeepSeek model equipped with a two-stage retriever and reservoir-specific tool APIs, enabling accurate, secure, and fully automated reservoir-management workflows. By combining the efficiency of two-stage retrieval, the technical prowess of DeepSeek, and the flexibility of tool-augmented RAG, KRDS represents a comprehensive solution to the unique challenges of intelligent reservoir management in the digital transformation era.

Framework

The general workflow of the KRDS is illustrated on Fig.1. It is obvious that the KRDS reservoir agent contains two main steps. The first component is the two-stage knowledge retrieval and recall framework, which, when prompted by a reservoir engineer's query (e.g., "Check the production history of A1 well"), efficiently retrieves the most relevant information from a vast document repository. This framework begins with a Recall Phase, which utilizes dense retrieval to quickly identify a broad set of candidate passages. Following this, a Sequential Phase employs a more refined reranking mechanism to precisely score and select the most pertinent documents. The output of this retrieval process is a highly ranked knowledge context that is then passed to the second main step, the Retrieval-augmented conversational DeepSeek LLM framework. This framework processes the retrieved knowledge context along with the original query. Within this step, a Mission Planning module breaks down the complex request into actionable sub-tasks. It then uses an Execution Tool to interact with various System Tools, such as databases for well logs, production histories, and core analyses. The DeepSeek LLM acts as the central reasoning engine, synthesizing the retrieved knowledge and the data gathered by the tools. Finally, it formulates a coherent and accurate answer, which can include visualizations like the production history curve shown in the diagram, and presents it to the user. This integrated workflow ensures that the KRDS agent can autonomously complete complex reservoir management tasks with both high accuracy and data security.

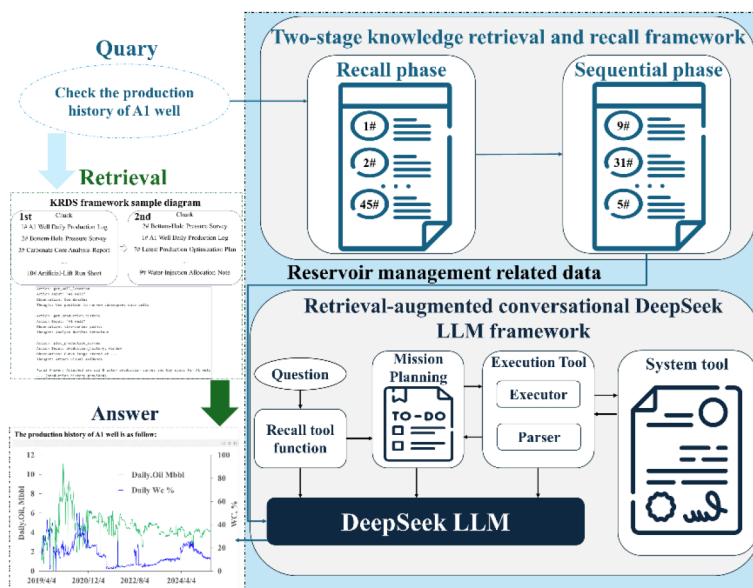


Figure 1—Specific workflow of the KRDS reservoir agent

Two-stage Retrieval of Reservoir Domain Knowledge

Fig.2 illustrates a sophisticated two-stage knowledge retrieval and ranking system designed for specific reservoir engineering queries, forming a core component of a Retrieval-Augmented Generation (RAG) framework. In the first stage, the Recall Phase, the system employs a Dense Passage Retrieval (DPR) model, such as the BCEmbedding-based dense retriever. This model encodes both the user's query and a vast repository of document passages into high-dimensional vectors. An Approximate Nearest Neighbor (ANN) search is then performed to rapidly identify the top 100 candidate passages whose vectors are most similar to the query vector, ensuring a high recall rate. This approach is highly efficient for sifting through large datasets.

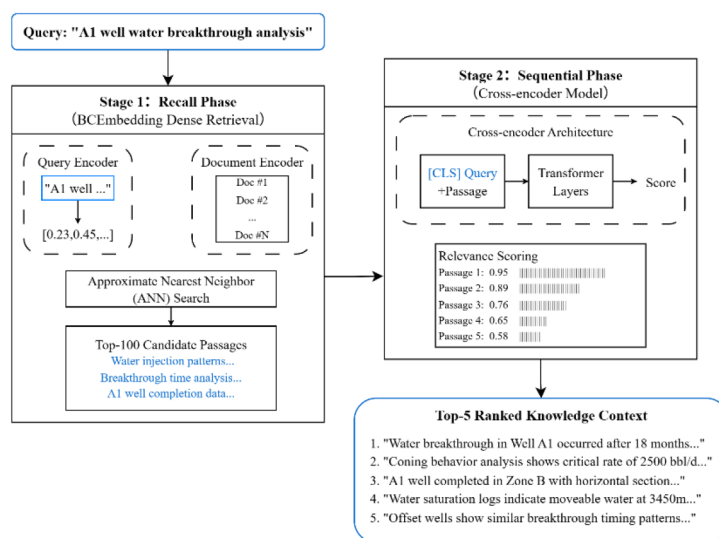


Figure 2—Two-stage Retrieval Pipeline for Reservoir Domain Knowledge

The second stage, the Sequential Phase, acts as a high-precision reranker. It utilizes a more computationally intensive cross-encoder model to jointly encode the user query and each of the 100 candidate passages. By analyzing the query-passage pair as a single input, this model gains a deep, contextual understanding of their relationship, allowing it to generate a precise relevance score for each one.

The system then outputs the top 5 highest-ranking passages as the "knowledge context." These passages, containing specific and factual insights on topics like "A1 well water breakthrough analysis," provide a robust data foundation for the AI agent to generate accurate, well-grounded, and contextually relevant responses, effectively mitigating the risk of hallucination in a domain where factual accuracy is critical.

Multi-turn Conversational Flow

Fig.3 provides a comprehensive illustration of the KRDS agent's workflow, showcasing its ability to handle a conversational sequence of queries that escalate in complexity. The process begins with a simple factual question, such as "What is the current production rate of A1 well?" The agent intelligently recognizes this intent, selects and executes a specific tool (`get_production_rate`), and provides a direct, data-driven answer. Building on this, when the user asks for a comparison to the historical average, the agent maintains the conversational context and executes a more complex tool chain, including `get_production_history` and `calculate_average`, to generate both a numerical summary and a detailed visualization of the production history. Finally, for an advanced predictive task like "Predict next month's production," the agent is triggered to perform a self-reflection, leading to the execution of a sophisticated tool chain (`decline_curve_analysis`, `production_forecast`) that includes uncertainty quantification, ultimately delivering a detailed forecast with a confidence interval. This multi-step process demonstrates how the KRDS agent effectively navigates from simple data retrieval to complex analytical and predictive tasks by intelligently leveraging its tool-use and context-aware capabilities.

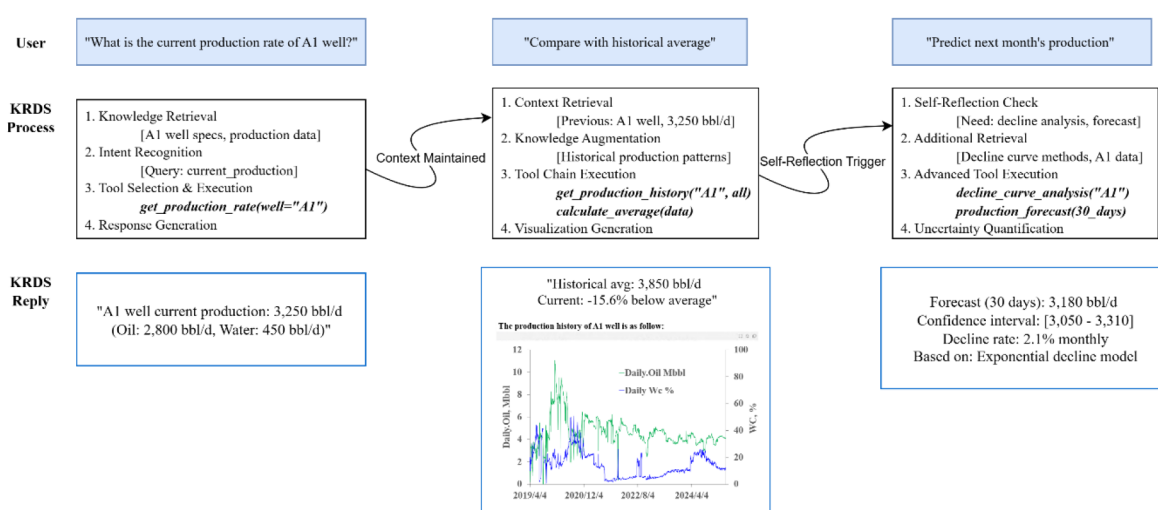


Figure 3—Multi-turn Conversational Flow with Tool Integration in KRDS

The Tool Classification and Execution Pipeline

A comprehensive tool execution pipeline used in reservoir engineering is shown in Fig.4, which integrates data retrieval, analysis, and prediction tools. The first step involves retrieving key reservoir data, such as porosity, permeability, pressure, saturation, production, and well trajectory, through specific data retrieval tools. Once the data is gathered, it undergoes various analyses like decline curve analysis, material balance, PVT analysis, nodal analysis, waterflood evaluation, economics calculation, and sensitivity analysis to assess reservoir performance and behavior. Based on the analysis, predictive tools such as production forecasting, water breakthrough prediction, enhanced oil recovery screening, reserve estimation, and risk assessment are applied to forecast future performance and identify risks. The entire process follows a structured pipeline that includes parsing user intent, matching tools to goals, extracting parameters, validating inputs, executing tool APIs, and formatting the response. This pipeline ensures a streamlined and efficient approach to managing and optimizing reservoir development.

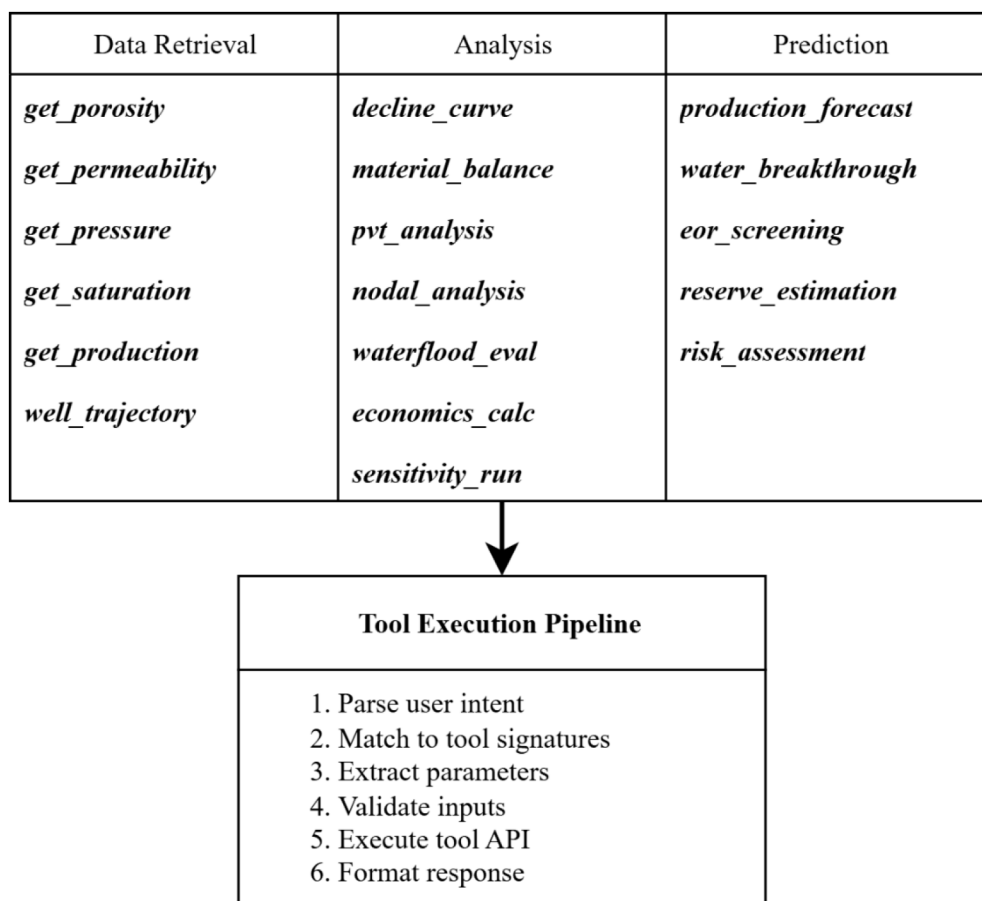


Figure 4—Reservoir Management Tool Classification and Execution Pipeline

Case study

Taking a specific oil reservoir block of Thamama formation from the Persian Gulf Basin in the Middle East as a case study, this paper tests the large-scale model of the scenario by selecting relevant real-world cases from the reservoir dynamics analysis process. The focus is on performing auxiliary dynamic analysis for a subset of injection-production well groups. This includes 18 specialized reservoir management functions, such as those for ‘permeability estimation,’ ‘porosity determination,’ ‘production rate calculation,’ and ‘wellbore trajectory analysis,’ among others. These functions are employed to address 45 scenario-based questions that reflect real-world operational practices, providing insights into the practical applications and challenges faced during daily field operations. The results of these analyses are critical for refining reservoir management strategies and optimizing production in complex field environments.

A comparison was conducted between four different combinations of reservoir automation agents: the KRDS, automated agent in the vertical domain, DeepSeekwithtraditionalvectorretrieval, DeepSeek under general knowledge base, as illustrated in Fig.5. The automation agent utilizing two-stage knowledge retrieval and DeepSeek invocation demonstrated the ability to effectively address domain-specific queries by considering various influencing factors. The two-stage knowledge retrieval model first filters domain-specific knowledge texts, identifying relevant professional knowledge related to the query. The large language model, through contextual learning, receives pertinent expertise from the vertical domain knowledge base and generates a comprehensive task plan along with the associated tool invocations. The success rate and the correct path rate for every hundred experiments are 83% and 87%, respectively, which extremely high than other three frameworks.

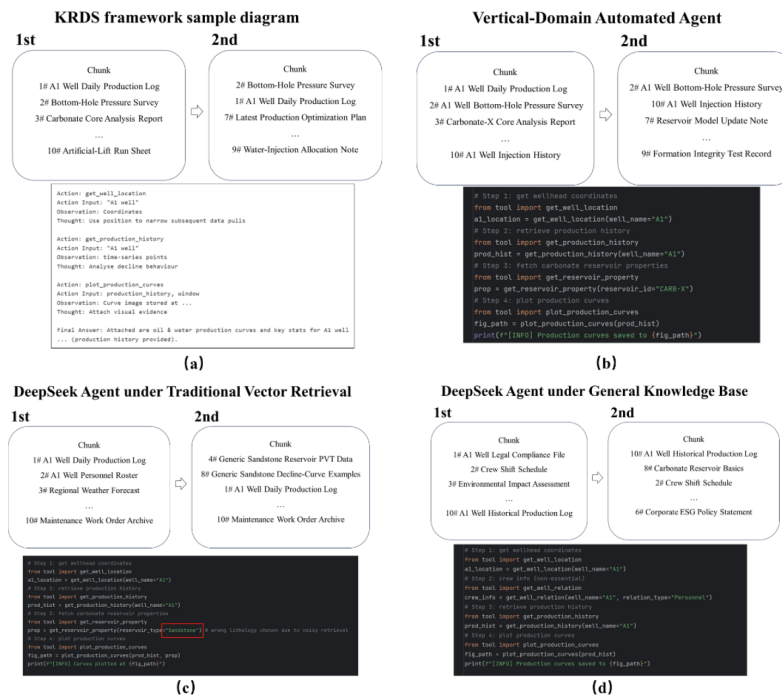


Figure 5—Diagram of comparison (a. KRDS; b. automated agent in the vertical domain; c. DeepSeek with traditional vector retrieval; d. DeepSeek under general knowledge base)

The reservoir dynamic analysis scenario model, KRDS, is designed to assist in querying and organizing historical data for any individual well, offering significant convenience for reservoir engineers to access and review data at any time. When the query is selected as "Check the production history of A1 well," the system follows a structured process to retrieve, analyze, and present the relevant information. The detailed workflow and the presentation of the results are illustrated in Fig.6. The model leverages the two-stage knowledge retrieval system to filter and extract the necessary historical data, which is then processed by the integrated DeepSeek LLM to ensure accurate and contextually relevant information is provided. The results are displayed in an intuitive format, allowing engineers to quickly assess the production history and make informed decisions based on the data retrieved. This capability not only streamlines the data retrieval process but also enhances operational efficiency, ensuring that engineers have easy access to critical historical performance data when making reservoir management decisions.

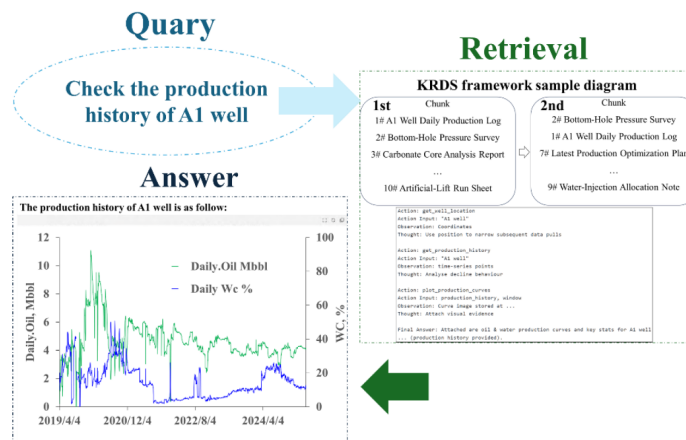


Figure 6—The specific process of result in the query of KRDS

Conclusion

In this study, we addressed the gap between the general capabilities of large language models (LLMs) and the specialized requirements of intelligent reservoir management, particularly in terms of domain expertise and data security. To bridge this gap, we proposed and implemented the Knowledge-Retrieval–DeepSeek System (KRDS), a tailored LLM-powered agent designed for reservoir engineering tasks. KRDS leverages a two-stage knowledge retrieval and recall framework, integrating eighteen specialized tool functions to autonomously execute a broad range of tasks from data retrieval to complex analysis and predictions. Our case study on a carbonate reservoir in the Persian Gulf Basin demonstrated the system's effectiveness, achieving an 83% success rate and an 87% correct path rate across forty-five operational scenarios. These results validate our approach of combining targeted knowledge retrieval with an on-premises, open-access LLM as a robust solution for intelligent reservoir management. KRDS is the first LLM agent tailored to the domain that not only executes intricate tasks but also adheres to strict enterprise-level data security standards. This research provides a foundational proof of concept, showcasing the potential of this framework to enhance operational efficiency and streamline oil and gas field development. Future work will expand the KRDS toolset to include advanced predictive models, multi-modal capabilities for interpreting seismic and geological data, and applications to other areas within the upstream sector.

Acknowledge

We acknowledge financial support from the National Science and Technology Major Project (2025ZD1406402), and the China Postdoctoral Science Foundation (2025M774390).

Reference

- Asai, A., Wu, Z., Wang, Y. et al 2023. Self-RAG: Learning to Retrieve, Generate, and Critique Through Self-Reflection. arXiv:2310.11511.
- Chang Y., Wang X., Wang J., et al A survey on evaluation of large language models[J]. *ACM transactions on intelligent systems and technology*, 2024, **15**(3): 1–45.
- DeepSeek-AI Team. 2024a. DeepSeek-V2: Multi-Head Latent Attention for Efficient LLMs. arXiv:2403.04163.
- DeepSeek-AI Team. 2024b. DeepSeek-Coder: An Open 67-Billion-Parameter Code LLM. arXiv:2403.09045.
- Karpukhin, V., Oguz, B., Min, S. et al 2020. Dense Passage Retrieval for Open-Domain Question Answering. Proc. EMNLP 2020, 6769–6781. DOI: [10.18653/v1/2020.emnlp-main.550](https://doi.org/10.18653/v1/2020.emnlp-main.550)
- Gao, L., Kim, L., Chen, M. et al 2025. LevelRAG: A Unified Framework for Retrieval-Augmented Large Language Models in Technical Domains. arXiv:2502.18139.
- Lewis, P., Perez, E., Piktus, A. et al 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *Adv. NeurIPS* **33**, 9459–9474.
- Liu H., Ren Y., Li X., et al Research status and prospects of application of artificial intelligence mega-models in the oil and gas industry[J]. *Petroleum Exploration and Development*, 2024, **51**(4): 910–923.
- Liu J., Chang H., Forrest J.Y.L., et al Influence of artificial intelligence on technological innovation: Evidence from the panel data of china's manufacturing sectors[J]. *Technological Forecasting and Social Change*, 2020, **158**: 120142.
- Liu, J., Wang, R., Chen, F. et al 2025. Quantitative Analysis of Performance Drop in DeepSeek Models Under Low-Bit Quantization. arXiv:2505.02390.
- Mohaghegh S.D. Reservoir simulation and modeling based on artificial intelligence and data mining (AI&DM)[J]. *Journal of Natural Gas Science and Engineering*, 2011, **3**(6): 697–705.
- NetEase Youdao. 2023. BCEmbedding: Bilingual and Cross-Lingual Embedding Framework. <https://github.com/netease-youdao/BCEmbedding>.
- Nogueira, R. and Cho, K. 2019. Passage Re-Ranking with BERT. arXiv:1901.04085.
- Parshall J. Mature fields hold big expansion opportunity[J]. *Journal of Petroleum Technology*, 2012, **64**(10): 52–58.
- Osco L.P., Wu Q., De Lemos E.L., et al The segment anything model (sam) for remote sensing applications: From zero to one shot[J]. *International Journal of Applied Earth Observation and Geoinformation*, 2023, **124**: 103540.
- Schick, T., Zoellner, M., Schütze, H. et al 2023. ToolFormer: Language Models Can Teach Themselves to Use Tools. arXiv:2302.04761.
- Wang L., Yao Y., Luo X., et al A critical review on intelligent optimization algorithms and surrogate models for conventional and unconventional reservoir production optimization[J]. *Fuel*, 2023, **350**: 128826.

- Wolf T., Debut L., Sanh V., et al Transformers: State-of-the-art natural language processing[C]//Proceedings of the 2020 conference on empirical methods in natural language processing: system demonstrations. 2020: 38–45.
- Yao, S., Zhao, H., Yu, D. et al 2023. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629.
- Zhang, H., Liu, J., Zhu, Z. et al 2024. Efficient and Effective Retrieval of Dense-Sparse Hybrid Vectors Using Graph-Based Approximate Nearest-Neighbor Search. arXiv:2410.20381.